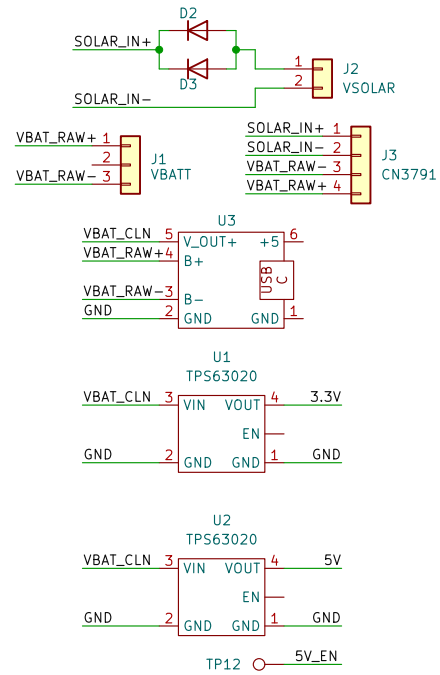


External Modules (Battery/power management)

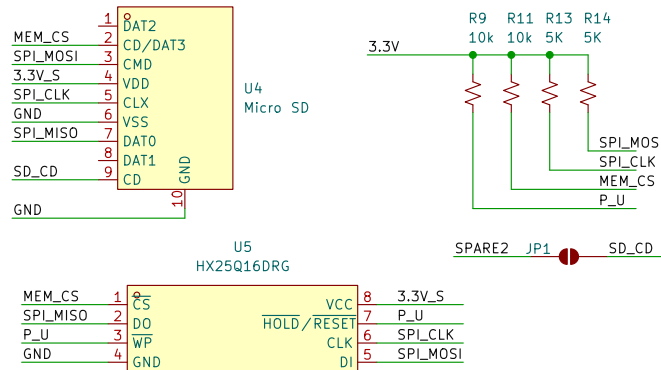
Panel -> CN3791 -> Battery -> TP4056 (for UVLO and USB charging) -> TPS63020 for main power (3.3V) + Aux 5V (SPS30)

TPS63020 enabled by default. External pulldown connection needed to disable them (only for aux 5V).



SD card & SPI Flash & related pull-up

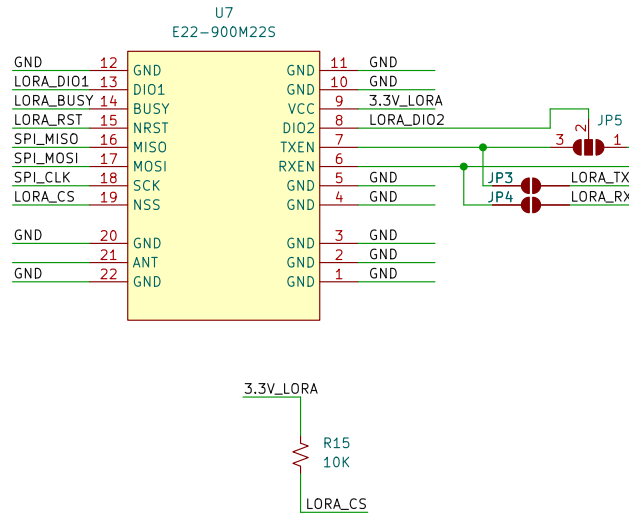
CS is shared between SD and SPI flash. They have been added to support further development later on, they are not supposed to be used together, you can install the SPI flash OR the SD card.



TP4056 board has a common positive battery disconnecter, which is not ideal for our use case. The only option is to leave the battery and the related charging circuit floating in respect to everything else (or add a separate UVLO). Keep it in mind while probing around.

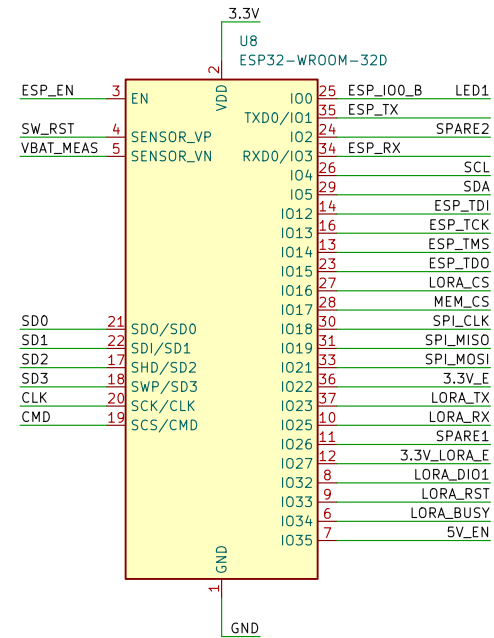
LORA Module

LORA_DIO2 MAY be used to automatically toggle between reception and transmission, hence the various jumpers to modify the connection based on what the library actually needs.



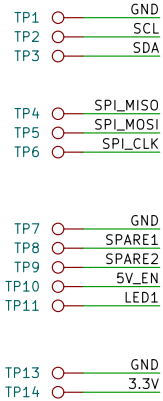
Main processor

ESP32 SoM



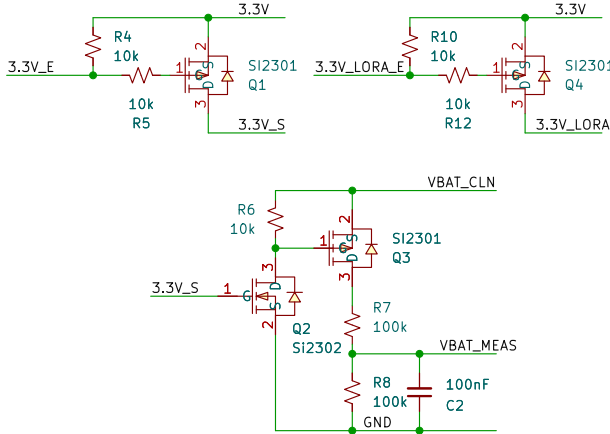
Test points

The beaded hooks for attaching the oscilloscope probe are cheap and plentiful, and always nice when stuff does not go as planned.



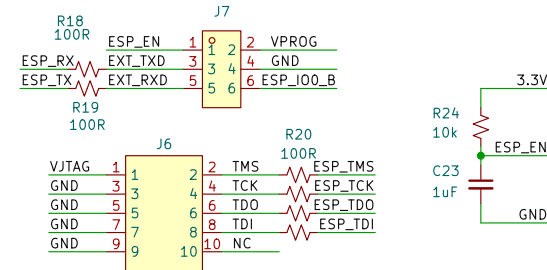
Power switching circuit

Remove power to the peripherals during standby. This includes the sensors, the I2C pullups resistor, and the voltage divider to measure the cell voltage.

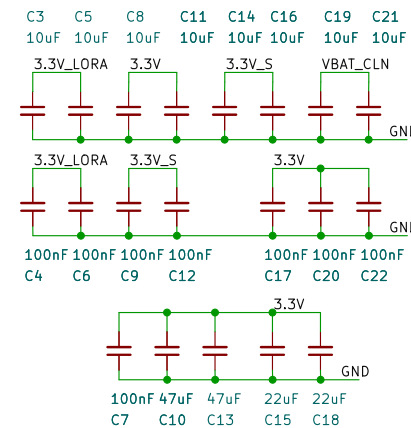


Programming Interfaces (JTAG/UART) & related

VPROG not connected to 3.3v. I dont want any power backfeeding in the regulator.

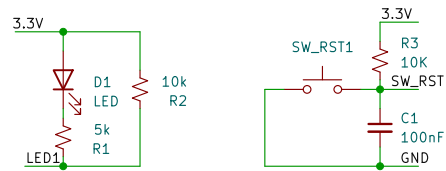


Decoupling capacitors



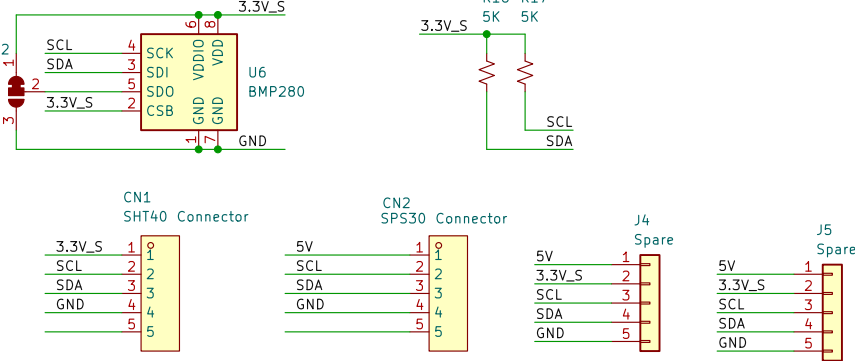
I/O for user interaction

Led connected to the pins used to reset the board during programming, as such this arrangement is needed. The switch is used to perform the factory reset of the board.



Sensors

Default address: 0x77 (0x76 if JP2 moved to GND)



Author: Andrea Cerfaglia
<https://thebigboschi.github.io/>

Sheet: /
 File: ESP32 Data Acquisition Board.kicad_sch

Title: ESP32 based data acquisition board

Size: A3
 Date: 2026-06-07
 KiCad E.D.A. 10.0.3

Rev: 0.1
 Id: 1/1